

EXERCISES

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UA4. DATA SETS COMBINATION

EXERCISE 1

The *mydata.sales88*, *mydata.sales89* and *mydata.sales90* data sets contain data on the sales of a company for the years 1988, 1989 and 1990, respectively. The three tables contain the variables *month* (character) and *sales* (numeric) and an observation for each month.

- a) Create a temporary data set named *sales1* corresponding to the concatenation of three data sets.
- b) In a single Data Step, create a temporary data set named *sales2* corresponding to the concatenation of three data sets; and create a character variable named *year* that takes the values 1988, 1989 and 1990, according to the data set that the observations belong.

EXERCISE 2

Consider the *mydata.employees* and *mydata.firm* data sets, the latter containing the names of the heads of each department.

- a) Create a data set named *employees1* that includes all employees of the company and respective department heads.
- b) Adapt the previous data step in order to create a variable (*Total*) that represents the accumulated total of the employee salaries by department.
- c) Exclude from the data set created in the previous paragraph the *City*, *Name*, *Birth* and *Total* variables; place the names of the heads of departments in uppercase and sort the data by the last names of employees. In the event of equal surnames then order it by *ID*. Use DATA STEP.
- d) Solve the previous item by using PROC SQL.

EXERCISE 3

Merge the *purchase* and *inventory* datasets to create a new temporary dataset (*pur_price*) where the price value found in the *inventory* dataset is added to each observation in the *purchase* dataset, based on the model number.

There are some templates in the *inventory* dataset that were not purchased (and therefore are not in the dataset *purchase*). Do not include these templates in the new dataset.

Based on the *Quantity* variable of the *purchase* dataset, calculate a total cost (*TotalCost*) equal to the *Quantity* times the *Price*.

EXERCISE 4

Import *card_base.csv* and *customer_base.csv* to SAS. These 2 datasets contain information about the type of card and credit limit (*card_base*) and about the customer age and segment (*customer_base*).

Create a new data set that only contains the customers with a credit card (using PROC SQL). The output dataset should only contain the *Cust_ID*, *Card_Family* and *Customer_Segment*, for only customers with age less than 30 years.

EXERCISE 5

Import the datasets *item_category*, *daily_sales1*, and *daily_sales2* to SAS. These 3 datasets contain sales data of vegetables in a supermarket.

- a) Create a new dataset named *daily_sales* that concatenates *daily_sales1* and *daily_sales2*.
- b) Create a new dataset named *all_daily_sales* that joins *daily_sales* and *item_category*.

In the new dataset, drop the variable *Category_Code* and create a new one, named *Total_Sales*, that corresponds to the multiplication between *Quantity_Sold* and *Unit_Selling_Price*.

- c) Create a new dataset named *all_daily_sales1* based on the previous one that meets the following conditions:
 - a. Only contains sales greater than 50 without discount;
 - b. Only contains the variables *Time*, *Date*, *Item_Name*, *Category_Name* and *Total_Sales*;
 - c. The variable *Total_Sales* appears with just 2 decimal cases.
- d) Create a new dataset named *all_daily_sales2* based on the *all_daily_sales* that meets the following conditions:
 - a. Create a new variable named *Sales_Level*, whose values are as follows:
 - i. If *Total_Sales* is lower than 3, then it takes the value "Low sale";
 - ii. If *Total_Sales* is higher than or equal to 3 but lower than 5, then it takes the value "Medium sale";
 - iii. If *Total_Sales* is higher than or equal to 5, then it takes the value "Big sale";
 - b. Only contains the variables *Time*, *Date*, *Item_Name*, *Category_Name*, *Total_Sales* and *Sales_Level*.